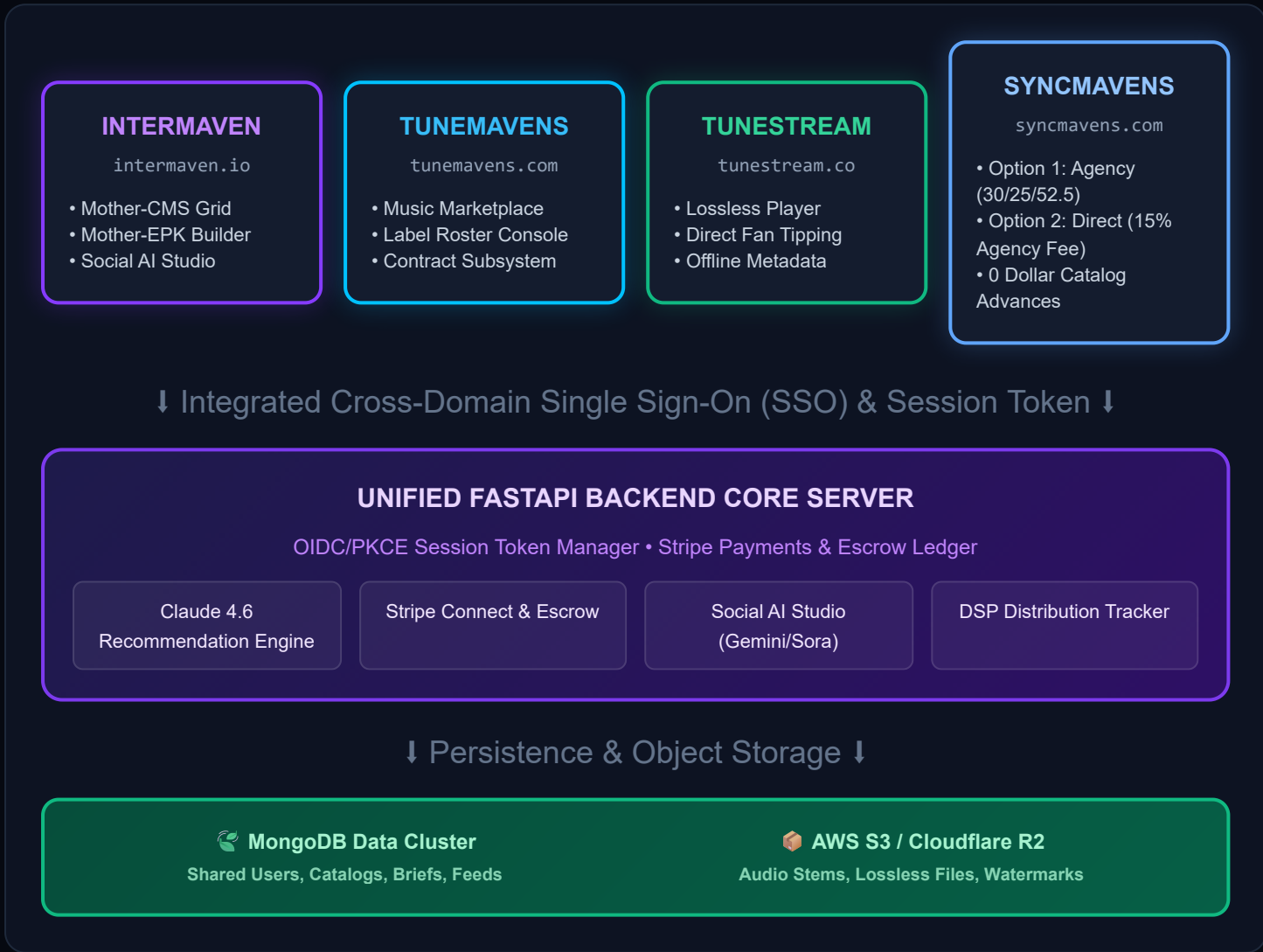


ECOSYSTEM SYSTEM ARCHITECTURE & FLOWCHART

Monorepo Technical Blueprint, Connectivity Flowchart & Data Schema Index

1. High-Impact Ecosystem Connectivity Flowchart



2. End-to-End Data & Asset Lifecycle Sequence

- Catalog Ingestion:** Creator uploads audio stems to TuneMavens. Stems store in AWS S3, metadata upserts to MongoDB catalogs with auto-generated ISRCs.
- Split Cascade Lock:** Creator defines split waterfall (90/10) in publishing_deals and signs contract via contracts.
- Sync Placement:** Music supervisor posts brief on SyncMavens. AI Match Simulator evaluates track compatibility. Pitch delivered with watermarked preview. Supervisor approves and licensing waterfall executes (Option 1: 30% Agency / 25% Admin / 52.5% Creator Pool vs Option 2: 15% SyncMavens Agency Fee / 85% Creator Pool).
- Consumer Streaming:** Track publishes to TuneStream. Fans stream lossless audio & tip artist directly. Revenue credits route to unified network balance.

5. **Social AI Marketing:** Creator triggers Intermaven Social AI Studio to render 1:1 and 9:16 creative assets (Gemini/Sora) and dispatch via Resend email / social DM.

4. Detailed Technical Stack Specifications & Monorepo Architecture

4.1 Monorepo Directory Layout & Application Boundaries

```
tunemaven/
├── apps/
│   ├── portal/           # Main Portal & Dashboard Hub (tunemavens.com)
│   ├── tunestream/       # Consumer Audio Streaming App (tunestream.co)
│   └── syncmavens/       # Sync Licensing Marketplace (syncmavens.com)
├── packages/
│   ├── shared-ui/        # Shared CSS Tokens, UI Components & Modals
│   └── shared-auth/      # Shared OIDC/PKCE Session Helpers
├── backend/
│   ├── server.py         # Primary FastAPI Application Entry
│   ├── auth.py           # PyJWT Cross-Domain Token Validation
│   ├── routes/           # Domain Routers (sso, match, stream, crm, cms)
│   └── models.py         # Pydantic v2 Data Transfer Models
└── deploy/              # Multi-Domain Hostinger VPS Nginx Configs
```

4.2 Layer-by-Layer Technical Specification Matrix

Tech Layer	Platform Standard	Implementation & Package Specification
Frontend Framework	React 18 + Vite	Modular Monorepo Monolith with Isolated Build Targets (dist/portal, dist/tunestream, dist/syncmavens).
Styling System	Vanilla CSS + Tokens	Custom HSL design tokens, glassmorphism, responsive grid layouts, Outfit/Inter fonts. Zero CSS framework bloat.
Native Packaging	Capacitor	Wraps web frontend applications for iOS & Android native deployment with native audio drivers.
Backend Engine	FastAPI (Python 3.11)	ASGI asynchronous web server running Uvicorn workers with non-blocking event loops.
Database Layer	MongoDB + Motor	Multi-tenant MongoDB replica set cluster using Motor for async Python database I/O.
Object Storage	AWS S3 / Cloudflare R2	Direct browser presigned URL upload pipeline for lossless audio stems (WAV/AIFF) and preview clips.
Single Sign-On	OIDC + PKCE	Custom OAuth2/OIDC SSO server (sso_router.py) issuing PyJWT cookies across .tunemavens.com.
Payment Gateway	Stripe Connect	Direct payouts, 4-tier entitlements, QR event ticketing scanner, Option 1 (30/25/52.5) & Option 2 (15% agency fee) ledgers.
AI Recommendation	Claude Sonnet 4.6	15s capped recommendation synthesis engine (users_router.py) recommending tools and apps.
Social AI Studio	Gemini Nano + Sora 2	Multi-format image (/generate-art) and short-form video clip (/generate-video) campaign generator.
CRM Engine	Resend API + Inbox	Multi-channel messaging engine dispatching emails via Resend and in-app inbox threads (messages).

